



RADIANT NUCLEAR • THERMAL MODELING INTERVIEW

3D CFD Conjugate Heat-Transfer Model of a Prismatic HTGR Unit Cell

Predicting peak TRISO fuel temperature under normal operation
— and proving the number is trustworthy.

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Helium-cooled microreactor (Kaleidos class) • OpenFOAM 7 CHT

STAR walkthrough

How I'll walk through this

A single engineering story, told in four moves.

S

One number decides everything: does the TRISO fuel stay under **1600 °C**? Peak fuel temperature is the licensing-relevant quantity.

T

The take-home: build a **3D CFD** model predicting peak fuel temperature (normal operation) — and **demonstrate validation**, not just plot a field.

A

A 3D conjugate heat-transfer model in OpenFOAM — unit-cell geometry, two coupled regions, an embedded fuel source, and a decisive **mesh-quality fix**.

R

670 ± 2 °C, grid-independent, corroborated three ways. ~930 °C of margin to the 1600 °C limit.

STAGE 01

SITUATION

The safety question that defines this reactor class — and the single number that answers it.



The safety case rests on peak fuel temperature

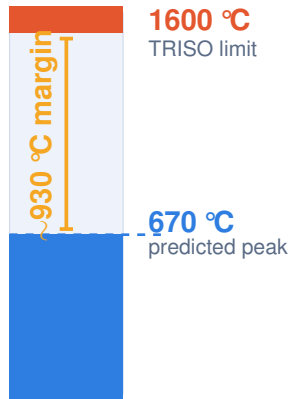
- Radiant's Kaleidos class is a **helium-cooled, TRISO-fueled** microreactor in a prismatic graphite block.
- **TRISO fuel** = fissile kernels in ceramic layers that retain fission products up to **~1600 °C**.
- Stay below that limit and the reactor is **safe by design** — so the whole thermal case reduces to one question.

The governing question

Does the fuel stay below **1600 °C** at full power — and how much **margin** is there?

A prediction is only useful with evidence it is correct.

Prismatic Microreactor Thermal Study • 3D CFD



Normal-operation headline result (this study).

STAGE 02

TASK

Turn that question into a well-posed, validated 3D CFD problem I could actually run and defend.



The take-home, and how I scoped it

What was asked

Build a **3D CFD** model to predict peak fuel temperature, and **demonstrate validation** — for the normal-operation scenario.

Scoping decisions

- **Representative unit cell**, not a whole block — standard prismatic-HTGR practice; tractable mesh, clean to validate.
- **Steady state** — only the equilibrium peak matters; transient-to-steady is infeasible (fluid CFL $\sim 10^{-4}$ s vs. minutes).
- Unit-cell **symmetry** $\Rightarrow$ adiabatic walls $\Rightarrow$ an *exact* energy balance.

Deliverable

A defensible peak-fuel-temperature prediction with a stated numerical uncertainty band.

Model $\rightarrow$ resolve the coolant flow + solid conduction together (conjugate heat transfer).

Validate $\rightarrow$ grid convergence, energy balance, and independent hand-calculations.

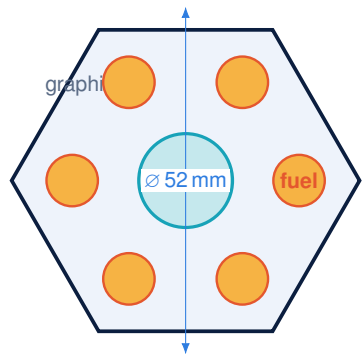
STAGE 03

ACTION

Geometry, physics, solver — and the mesh-quality problem that quietly moved the answer 87 °C.



Action 1 — The unit-cell geometry



Cross-section (extruded 0.8 m axially).

One hexagonal graphite prism, a central helium coolant channel, and a ring of six TRISO fuel compacts — built parametrically in **gmsh** (OpenCASCADE), fragmented so all regions share conforming faces, then extruded.

Parameter	Value
Hex circumradius	30 mm
Coolant channel dia.	16 mm
Fuel compact dia. ($\times 6$)	12.4 mm
Fuel-compact ring radius	18.8 mm
Active height (axial)	0.80 m

3 volumes $\rightarrow$ cellZones via `gmshToFoam`.

Action 2 — Physics & solver setup

Solver: steady `chtMultiRegionFoam` (OpenFOAM 7) — conjugate heat transfer across a coupled thermal interface.

- **Two regions only:** `fluid` (helium, compressible ideal gas) and `solid` (graphite + fuel).
- **Fuel** = a `topoSet` heat-source `cellZone` inside the solid, driven by `fvOptions` — the standard way to embed a volumetric fission source and avoid a fragmented mesh.
- **Turbulence:** $k-\varepsilon$ with wall functions (channel $Re \approx 1.2 \times 10^4$, turbulent).
- **Boundaries:** adiabatic outer wall + ends (symmetry); coupled fluid↔solid interface.

Operating point

Helium pressure	3 MPa
Inlet temperature	300 °C
Bulk velocity	~ 10 m/s
Reynolds number	$\sim 1.2 \times 10^4$
Fuel power density q'''	7 MW/m ³

✓ Internal sanity checks passed on the first converged run: helium density **2.52 kg/m^3** and fuel→wall conduction $\Delta T \approx 11 \text{ °C}$ both matched hand-calcs.

Action 3 — The mesh-quality problem (and the fix)

The first mesh used **triangular** prisms; triangles on the curved channel wall drove **max non-orthogonality** $\sim 86^\circ$, degrading the conduction (Laplacian) term.

Metric	Tri-prism	Hex (fix)	Target
Cell type	100% prism	100% hex	—
Non-orth. max (fluid)	86.0°	15.2°	<70°
Non-orth. avg (fluid)	61.8°	3.5°	<30°
Non-orth. max (solid)	86.0°	36.9°	<70°
Non-orth. avg (solid)	57.6°	10.1°	<30°

Max skewness 0.26 $\rightarrow$ 0.66 (target <4); both regions stay a single connected mesh.

This was not cosmetic. The near-wall-resolved hex mesh shifted the predicted peak fuel temperature **581 °C $\rightarrow$ 668 °C (+87 °C)** — direct evidence that resolving the film ΔT matters.

The fix (ADR-17)

Recombine to quads $\rightarrow$ extrude to **hexahedra**, plus **graded near-wall refinement** (targets $y^+ \approx 30$).

gmsh's structured BL field can't touch the conformal internal coolant wall; `snappyHexMesh addLayers` is the route to true inflation layers.

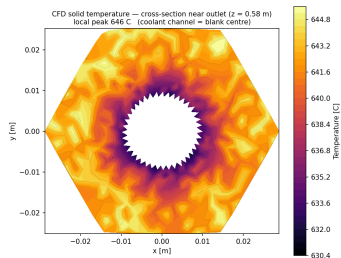
STAGE 04

RESULT

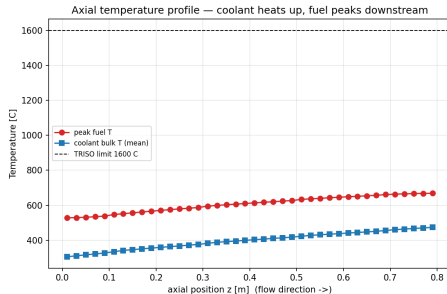
A grid-independent peak fuel temperature — corroborated three independent ways.



Result 1 — Peak fuel temperature



Solid temperature near outlet. Coolant channel = blank centre; hottest graphite sits between fuel compacts.



Fuel peaks downstream as the coolant heats up; both stay far below the 1600 °C limit.

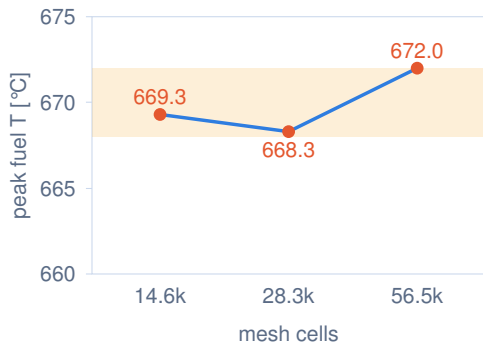
Peak fuel temperature = 670 ± 2 °C —
~930 °C margin to the 1600 °C TRISO limit.

Result 2 — Solution verification (grid convergence)

A prediction needs a **numerical uncertainty band**. I ran the normal case on three systematically refined hex meshes:

Hex mesh	Cells	Peak fuel T
Coarse	14.6k	669.3 °C
Medium (production)	28.3k	668.3 °C
Fine	56.5k	672.0 °C

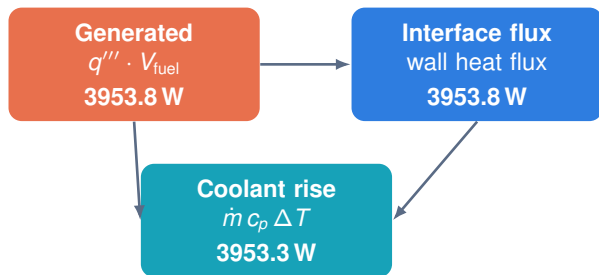
- Grid-independent to ± 2 °C (0.56%) across a $4\times$ cell range.
- Convergence is oscillatory (numerical noise floor), so I report the oscillation amplitude as the uncertainty.



The 66 °C gap vs. a coarse tri-prism (Richardson 604 °C) is that mesh smearing the near-wall film ΔT — the hex value is the trustworthy one.

Result 3 — Validation (energy & mass balance)

On the converged solution, three *independent* measures of the heat leaving the cell must agree if the physics is right. They close to **0.01%**.



Generated = crosses the wall = carried out by the coolant.

The adiabatic unit-cell symmetry makes this balance *exact* by construction — all generated heat has to leave through the coolant, so closure to 0.01% is a strong correctness check.

- **Mass** conserves to 0.000% ($\dot{m} = 5.05$ g/s, in = out).
- Coolant bulk $\Delta T = 150.8$ K (outlet 451 °C) — matches the ~ 154 K hand-calc.
- OpenFOAM function objects ($T_{\text{out}} = 723.83$ K, $\dot{m} = 5.047$ g/s) match the independent Python field parser exactly.

Result — The headline, in four numbers

670 °C

peak fuel T ($\pm 2^\circ\text{C}$)

~930 °C

margin to TRISO limit

0.01 %

energy-balance closure

0.56 %

grid-independence band

A **grid-independent, energy-balanced** peak fuel temperature with a **stated uncertainty band** — the evidence chain a reviewer of a thermal safety analysis actually looks for, produced by a from-scratch 3D CFD model.

Scope, limitations & next steps

🔍 Honest limitations

- Unit-cell **symmetry** $\Rightarrow$ adiabatic outer walls; block-edge and inter-cell effects not resolved.
- $y^+ \approx 19$ sits in the buffer/log range — slightly below the ideal wall-function band; aspect ratio 28–37 from long axial cells.
- No fully structured inflation layer (gmsh limitation); graded refinement captures most of the benefit.
- Steady state — absolute values are indicative given representative (non-vendor) material data.

➤ Natural next steps

- **snappyHexMesh addLayers** for true wall-normal inflation layers and a cleaner y^+ .
- **FEM↔CFD cross-check**: compare against my 2D finite-element conduction solver (element vs. volume, wall- h vs. resolved flow).
- **Loss-of-forced-cooling** case in 3D + a **transient** run with an ANS-5.1 decay-heat curve — the walk-away-safe story, time-accurate.

Every limitation above is stated, bounded, and has a defined path to close it.



THANK YOU

A validated 3D peak-fuel-temperature prediction — and the reasoning behind every number.

■ unit-cell chtMultiRegionFoam ■ hex mesh fix (+87 °C)
✓ 670 ± 2 °C peak 0.01% energy closure grid-independent